

SMARTX: Porewater SO₄/Cl

May 2026 Samples 333-463

2026-08-18

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##Add Required Packages

0.1 Run Information

```
##### Run information - PLEASE CHANGE
Date_Run = "2026-08-17" #Date that instrument was run
Run_by = "Victoria Ellis" #Instrument user
Script_run_by = "Victoria Ellis" #Code user
sample_year <- "2026"
sample_month <- "May"
Site = "GCREW"
run_notes = "One check standard is out of range" #any notes from the run

##### File Names - PLEASE CHANGE
#file path and name for raw summary data file
raw_file_name_cl = "Raw Data/SMARTX_202605_333-463_C1.txt"
raw_file_name_so4 = "Raw Data/SMARTX_202605_333-463_S04.txt"

#file path and name of processed data file
processed_file_name = "Processed Data/GCReW_SMARTX_PW_Processed_C1_S04_202605_333-463.csv"

##### Log Files - PLEASE CHECK
#downloaded metadata csv - downloaded from Google drive as csv for this year
Raw_Metadata = here("..", "GCReW", "GCReW_Project_Treatment_Metadata.csv")

#qaqc log file path for this year
Log_path = here("Porewater QAQC Logs", "GCReW_S04_C1_QAQC_Log.csv")
```

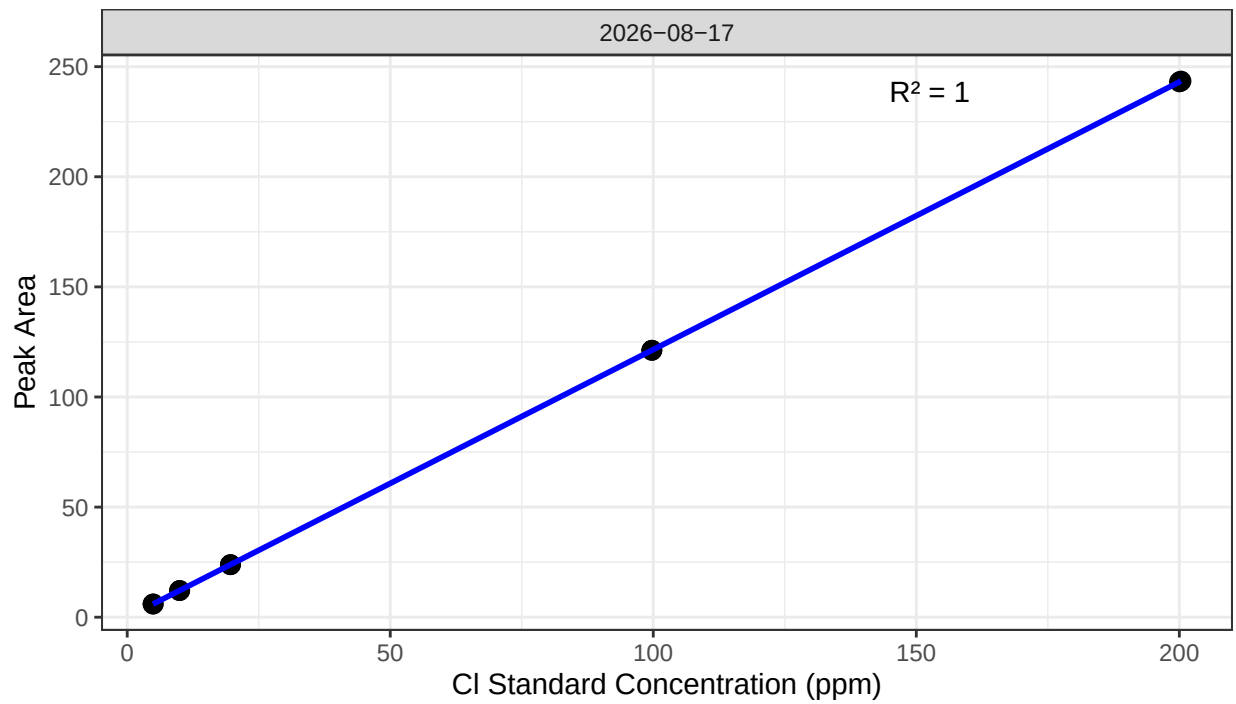
##Set Up Code - constants and QAQC cutoffs

##Read in metadata and create similar sample IDs for matching to samples

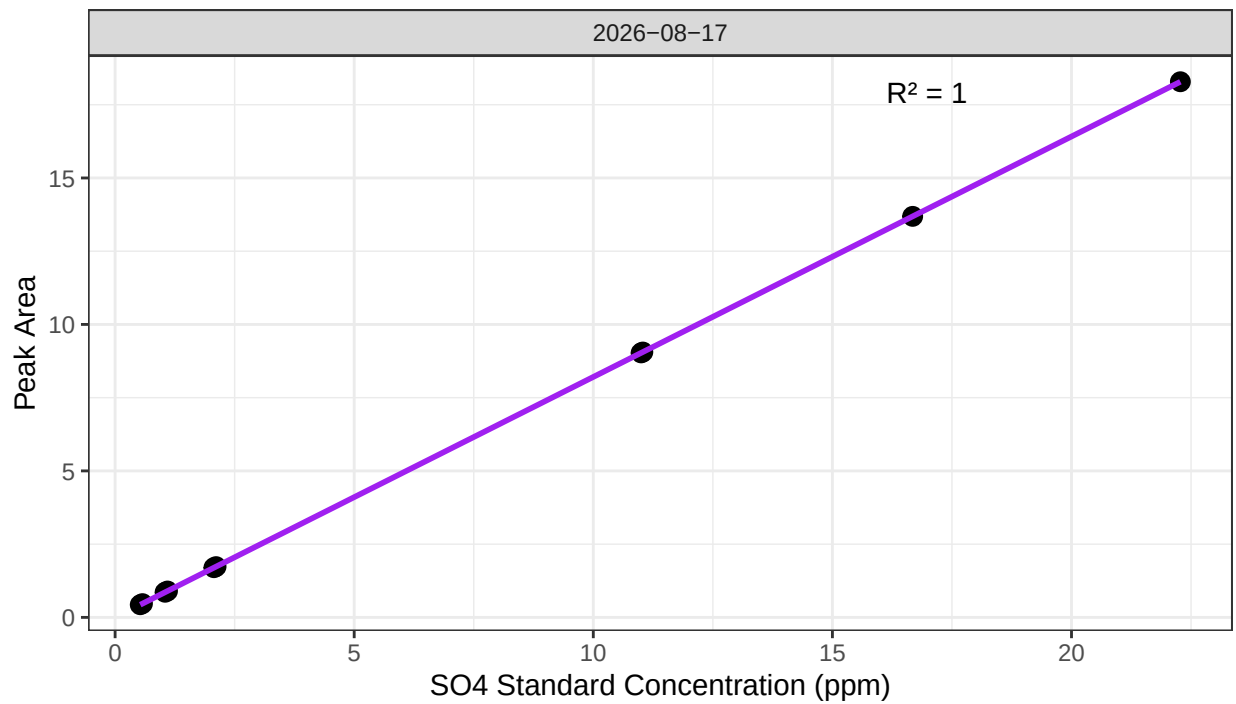
##Import Sample Data

0.2 Assess Standard Curves

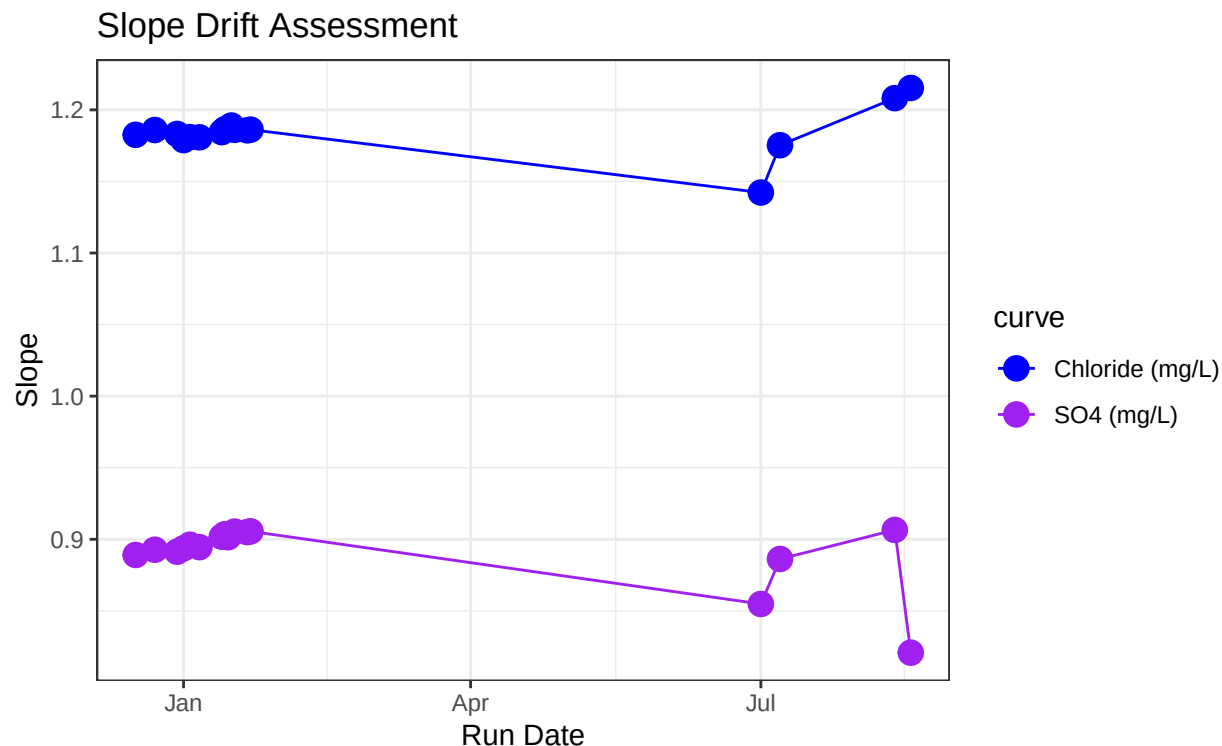
Chloride Std Curve



Sulfate Std Curve



```
## [1] "QAQC log file exists and has been read into the code."
```



```
## [1] "Cl Curve r2 GOOD"
```

```
## [1] "SO4 Curve r2 GOOD"
```

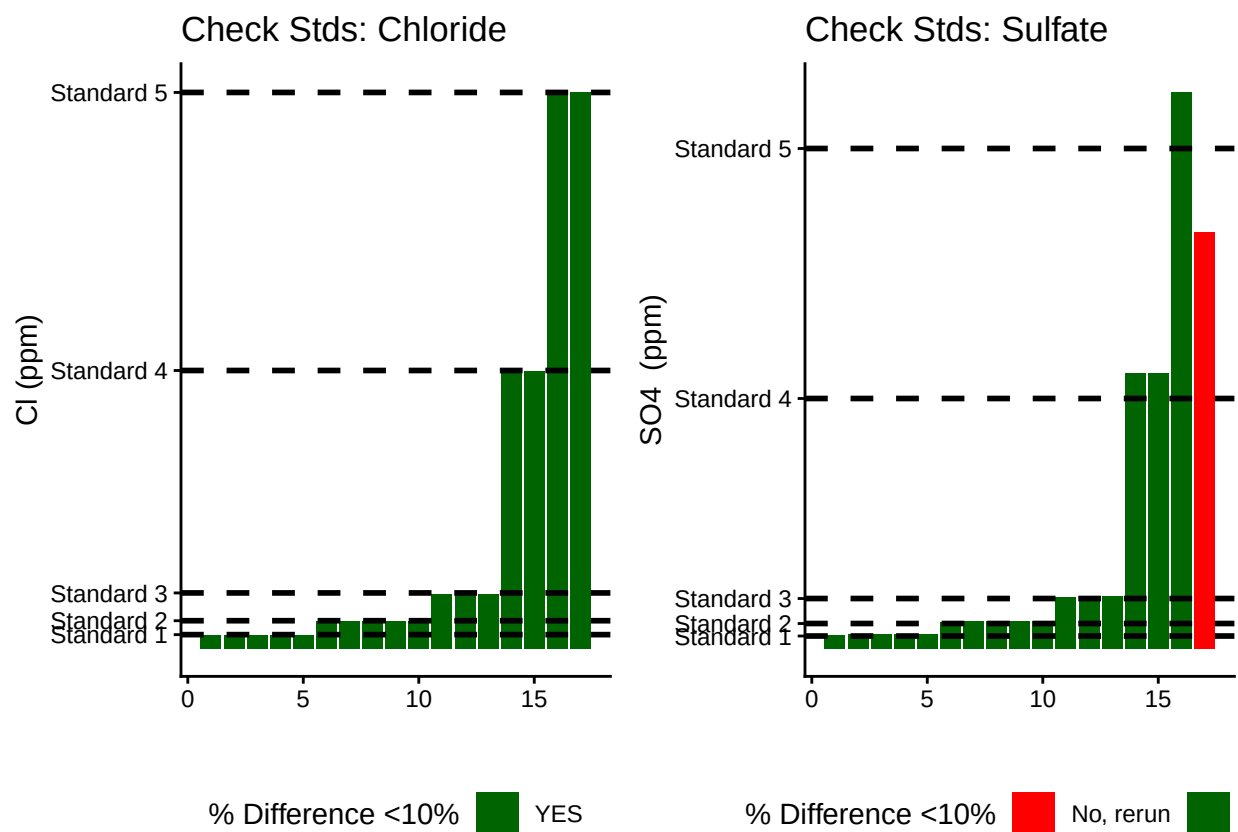
0.3 Assess Check Standards

```
## # A tibble: 5 x 5
##   sample_ID mean_Cl sd_Cl cv_Cl flag_Cl
##   <chr>      <dbl> <dbl> <dbl> <chr>
## 1 Standard 1  4.94 0.0224 0.00453 Chloride Check Standard RSD within Range~
## 2 Standard 2  9.97 0.0167 0.00167 Chloride Check Standard RSD within Range~
## 3 Standard 3 19.6 0.0228 0.00116 Chloride Check Standard RSD within Range~
## 4 Standard 4 99.7 0.00686 0.0000688 Chloride Check Standard RSD within Range~
## 5 Standard 5 200. 0.199 0.000994 Chloride Check Standard RSD within Range~
```

```
## # A tibble: 5 x 5
##   sample_ID mean_S04 sd_S04 cv_S04 flag_S04
##   <chr>      <dbl> <dbl> <dbl> <chr>
## 1 Standard 1  0.562 0.0201 0.0359 Sulfate Check Standard RSD within Range - ~
## 2 Standard 2  1.08 0.0215 0.0198 Sulfate Check Standard RSD within Range - ~
## 3 Standard 3  2.10 0.0263 0.0125 Sulfate Check Standard RSD within Range - ~
## 4 Standard 4 11.0 0.0250 0.00227 Sulfate Check Standard RSD within Range - ~
## 5 Standard 5 19.5 3.96 0.203 Sulfate Check Standard RSD within Range - ~
```

```
## [1] ">80% of Chloride Check Standards have RSD within range - PROCEED"
```

```
## [1] ">80% of Sulfate Check Standards have RSD within range - PROCEED"
```



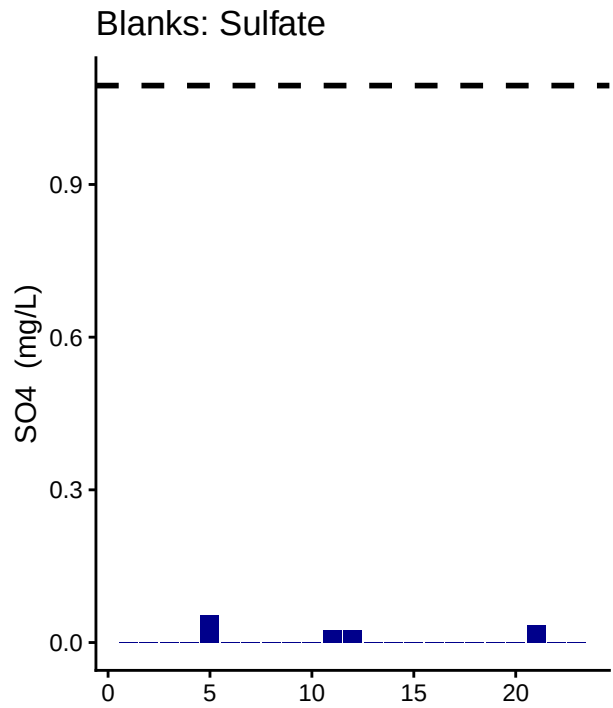
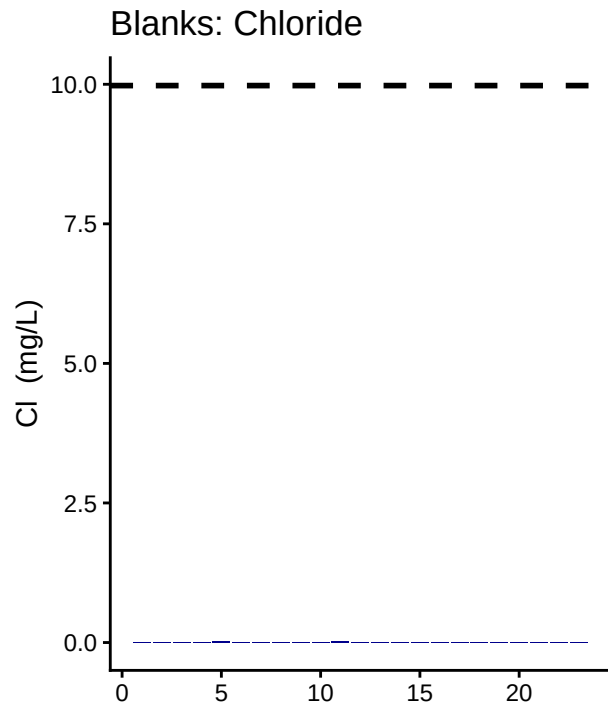
```
## [1] ">80% of Chloride Check Standards are within range of expected concentration - PROCEED"
```

```
## [1] ">80% of Sulfate Check Standards are within range of expected concentration - PROCEED"
```

0.4 Assess Blanks

```
## [1] ">80% of Chloride Blank concentrations are lower 25% quartile of samples"
```

```
## [1] ">80% of Sulfate Blank concentrations are lower 25% quartile of samples"
```



Blank Conc <25% Quartile Samples ☒ Y

Blank Conc <25% Quartile Samples ☒ Y

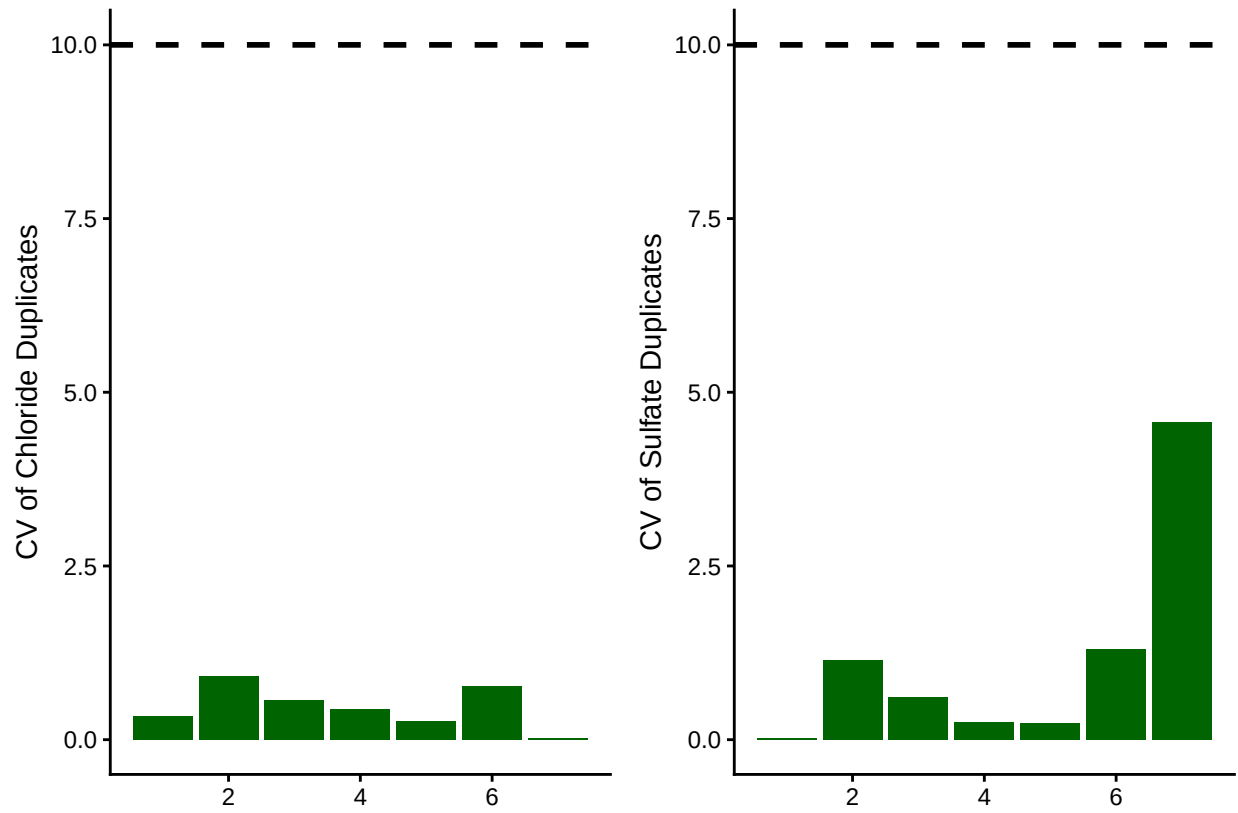
```
## Chloride blanks mean ppm:
```

```
## [1] 0.003021739
```

```
## Sulfate blanks mean ppm:
```

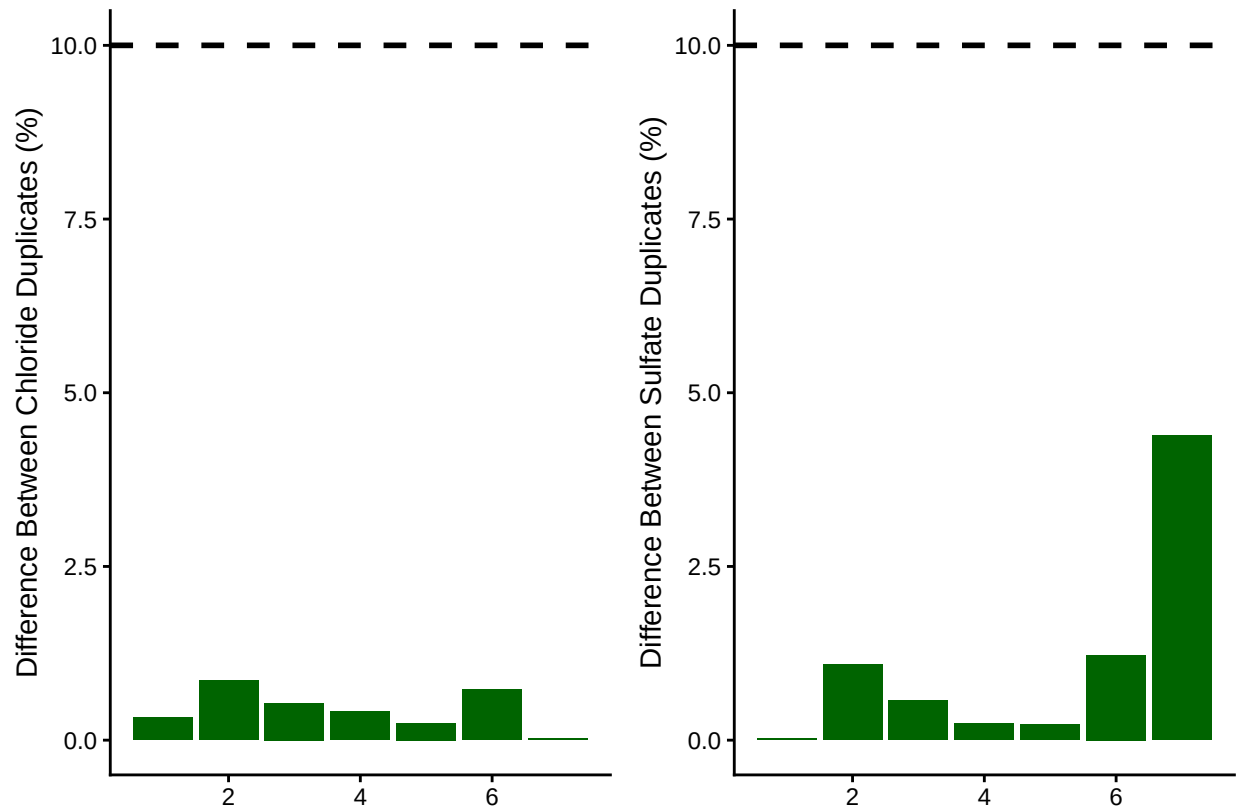
```
## [1] 0.005904348
```

0.5 Assess Duplicates



```
## [1] ">80% of Chloride Duplicates have a CV <10% - PROCEED"
```

```
## [1] ">80% of Sulfate Duplicates have a CV <10% - PROCEED"
```



```
## [1] ">80% of Chloride Duplicates have a percent difference <10% - PROCEED"
```

```
## [1] ">80% of Sulfate Duplicates have a percent difference <10% - PROCEED"
```

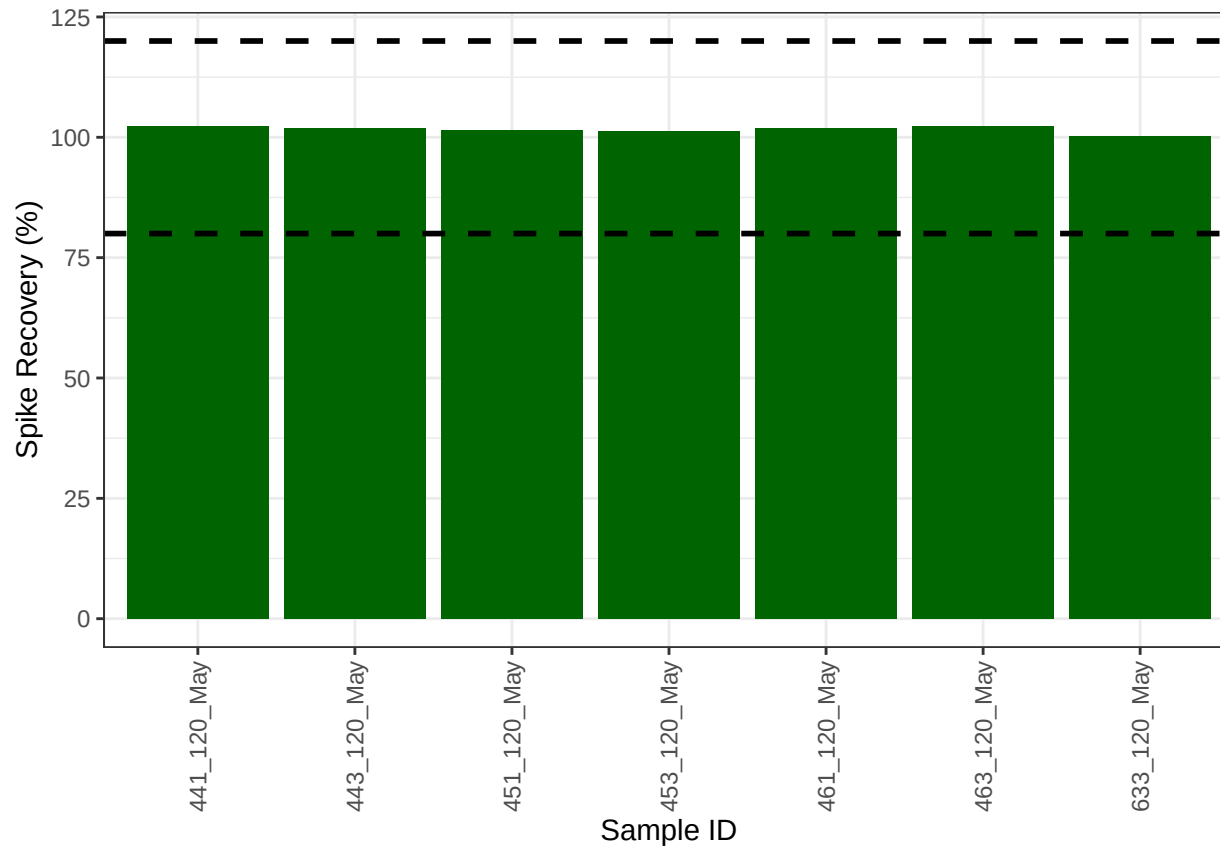
0.6 Calculate mmol/L concentrations & salinity, add dilutions

```
# Convert ppm to mmol/L
all_dat$S04_Conc_mM <- (all_dat$S04_ppm / s_mw)
all_dat$Cl_Conc_mM <- (all_dat$Cl_ppm / cl_mw)

# Calculate Salinity
# calculated using the Knudsen equation
# Salinity = 0.03 + 1.8050 * Chlorinity
# Ref: A Practical Handbook of Seawater Analysis by Strickland & Parsons (P. 11)
# = ((1.807*Cl_ppm)+0.026)/1000
all_dat$salinity <- ((1.8070 * all_dat$Cl_ppm) + 0.026) / 1000

#Need to determine dilution factors for your samples
all_dat$Dilution <- Dilution
# head(all_dat)
```


0.7 Assess Analytical Spikes



```
## [1] ">80% of S04 spikes have a recovery between the high and low cutoff - PROCEED"
```

0.8 Check if samples within the range of the standard curve

```
## Sample Flagging
```

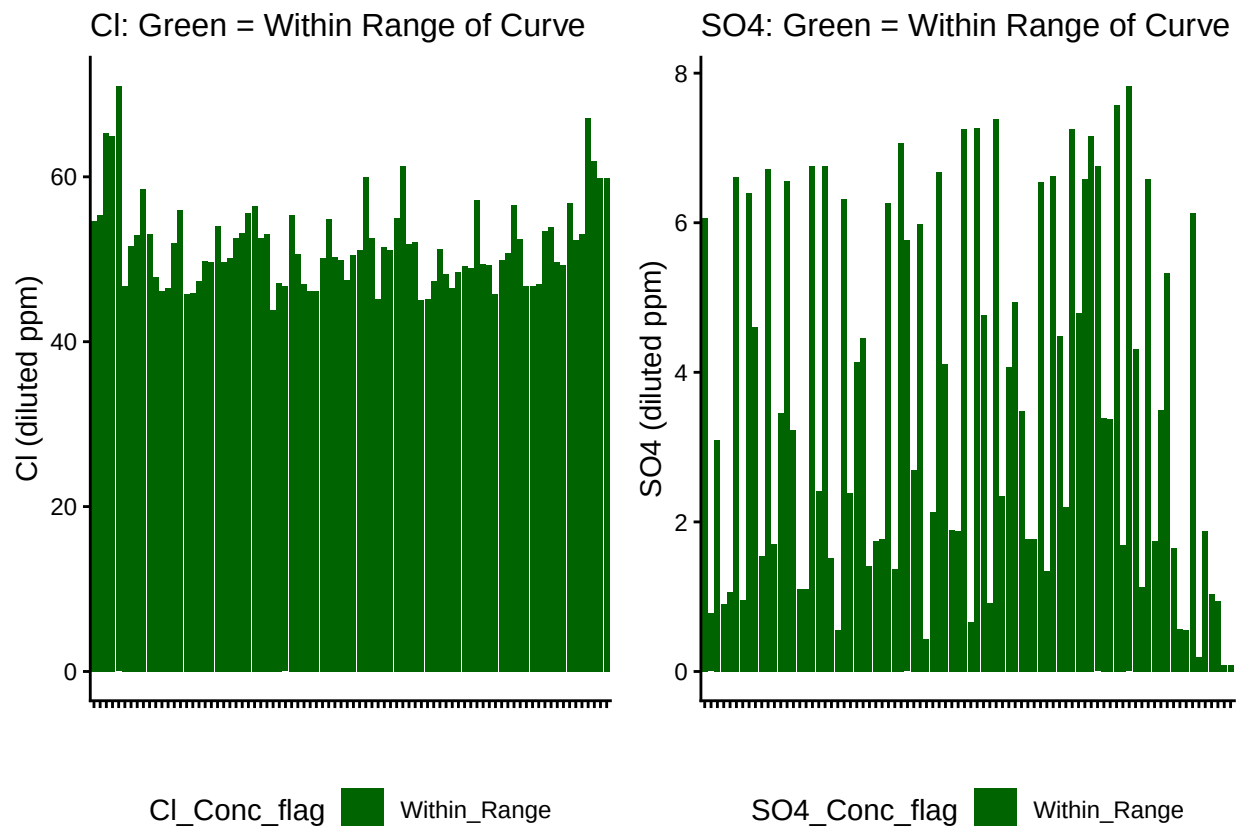


Table 1: SO4 samples

SO4_Conc_flag	Percent_samples
Within_Range	100

Table 2: Cl samples

Cl_Conc_flag	Percent_samples
Within_Range	100

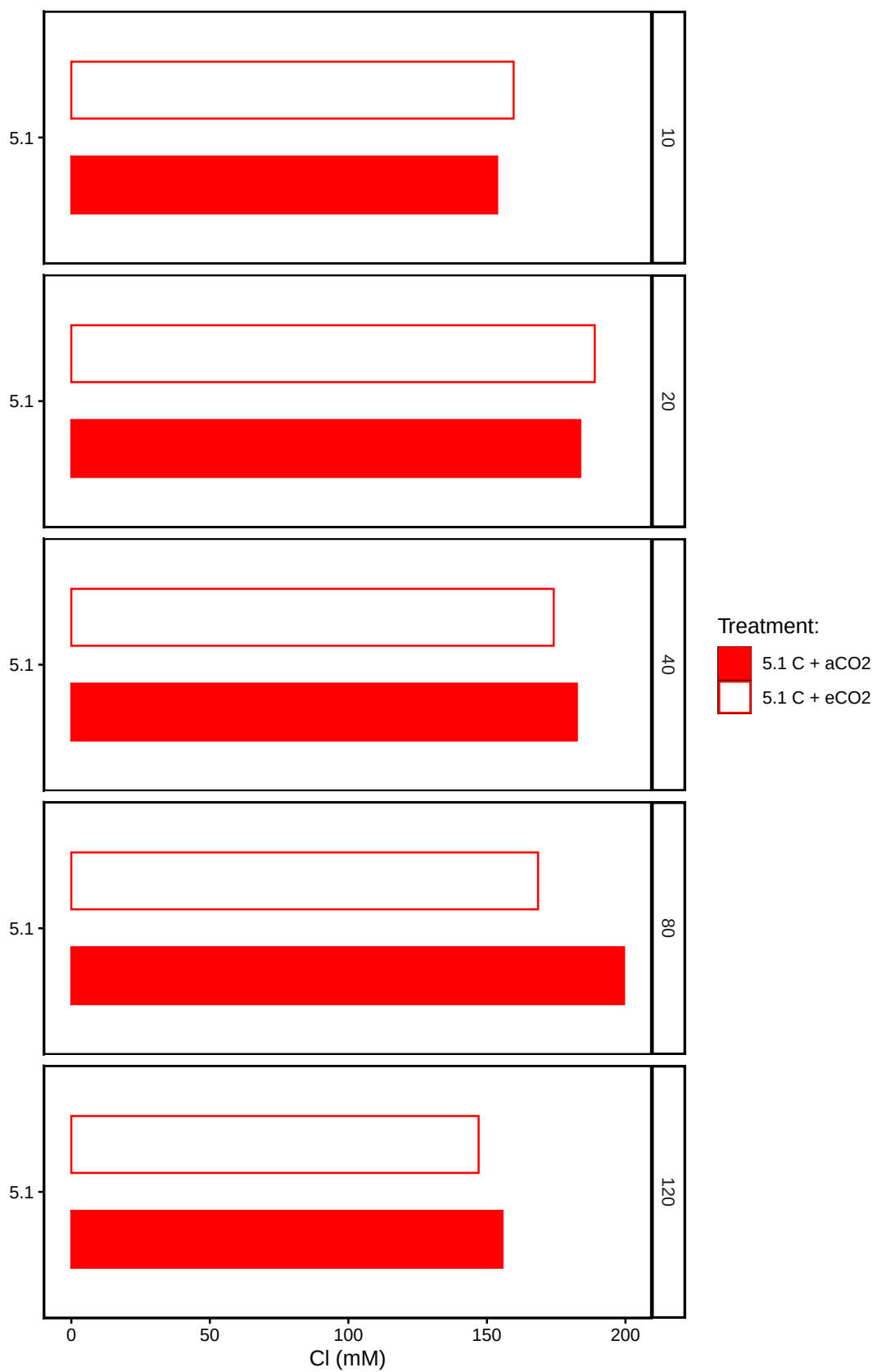
0.9 Check to see if samples run match metadata & merge info

```
## All sample IDs are present in metadata.
```

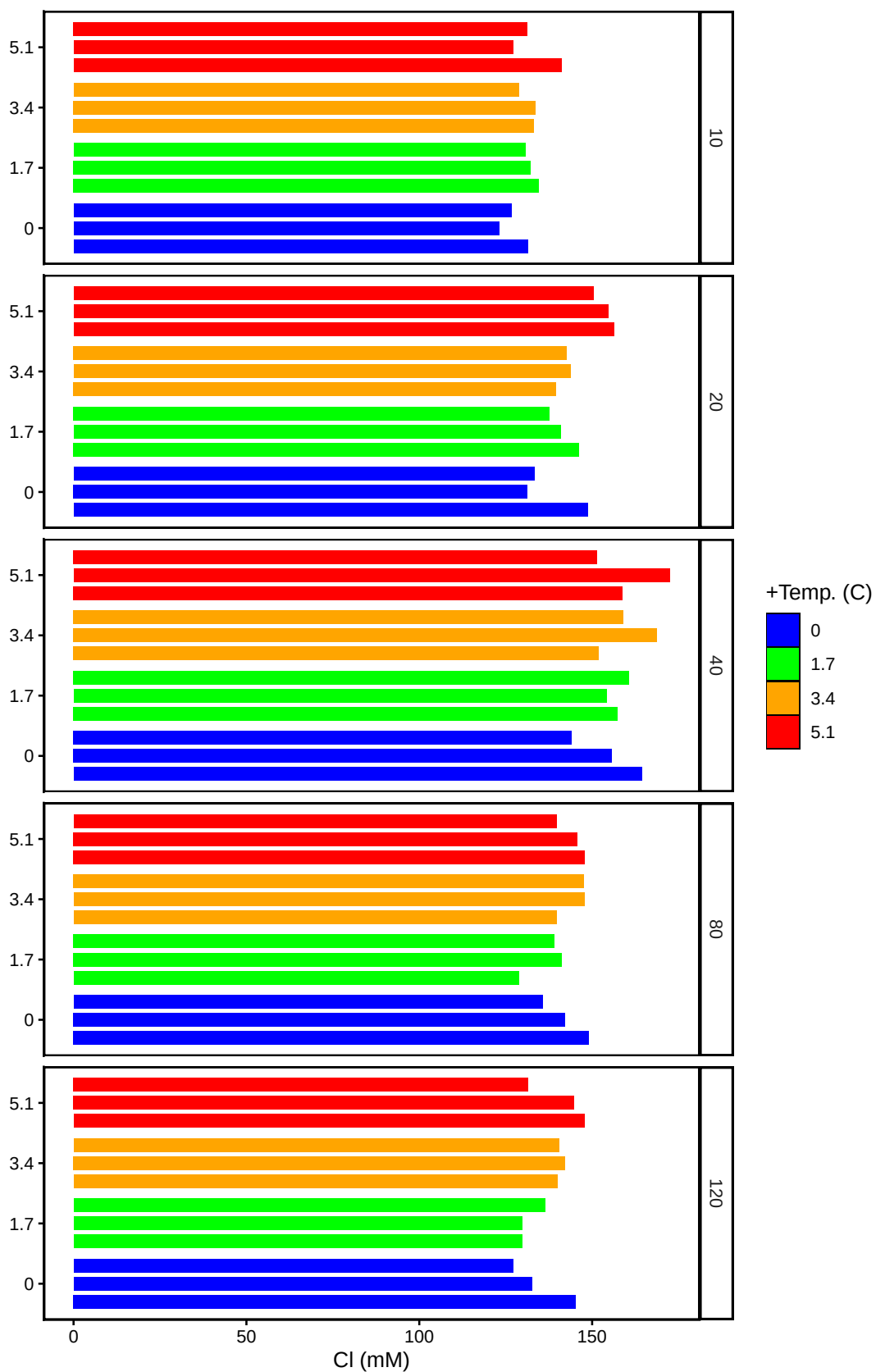
0.10 Visualize Data - Cl

```
## Visualize Data - Cl
```

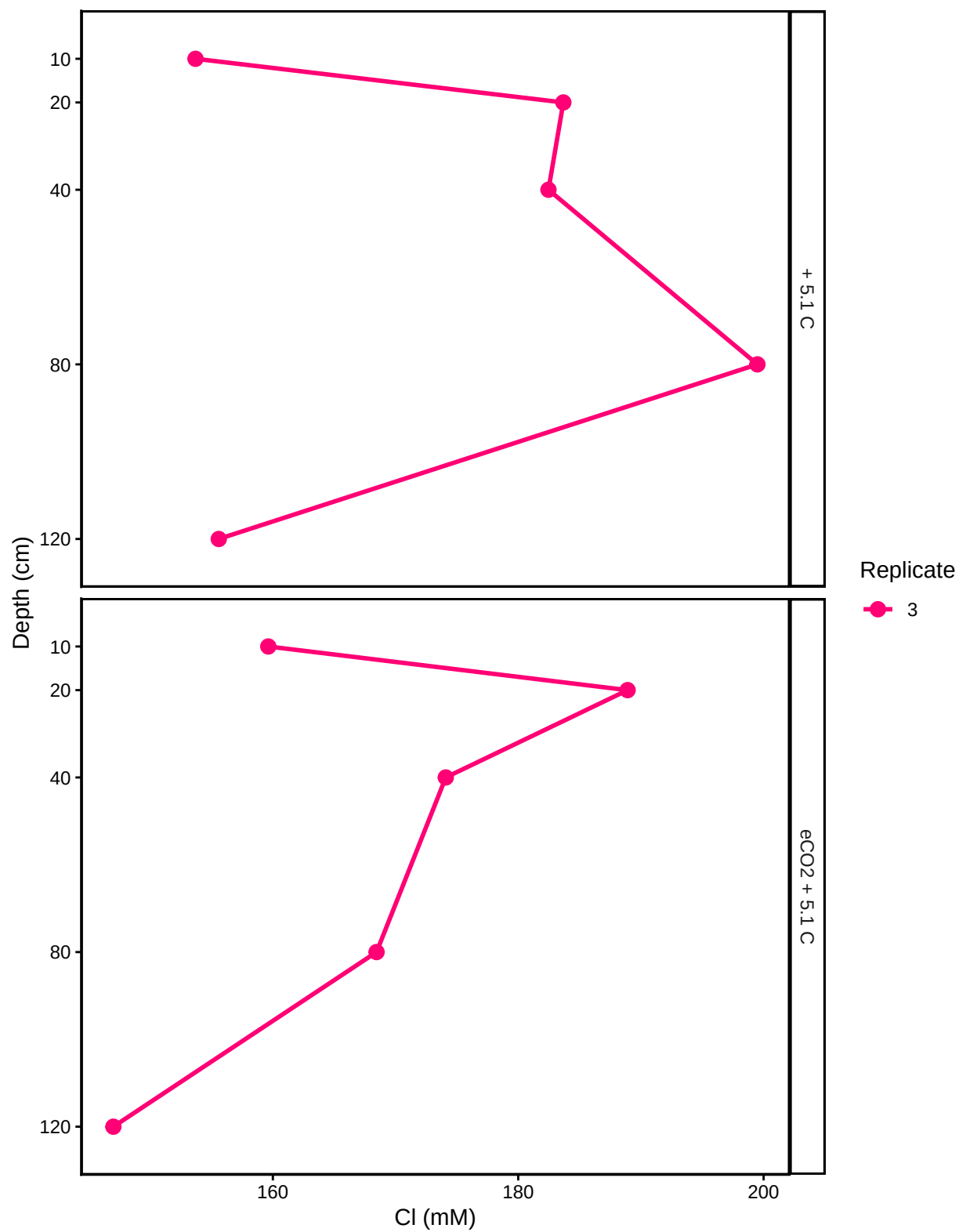
C3: Porewater Cl



C4: Porewater Cl

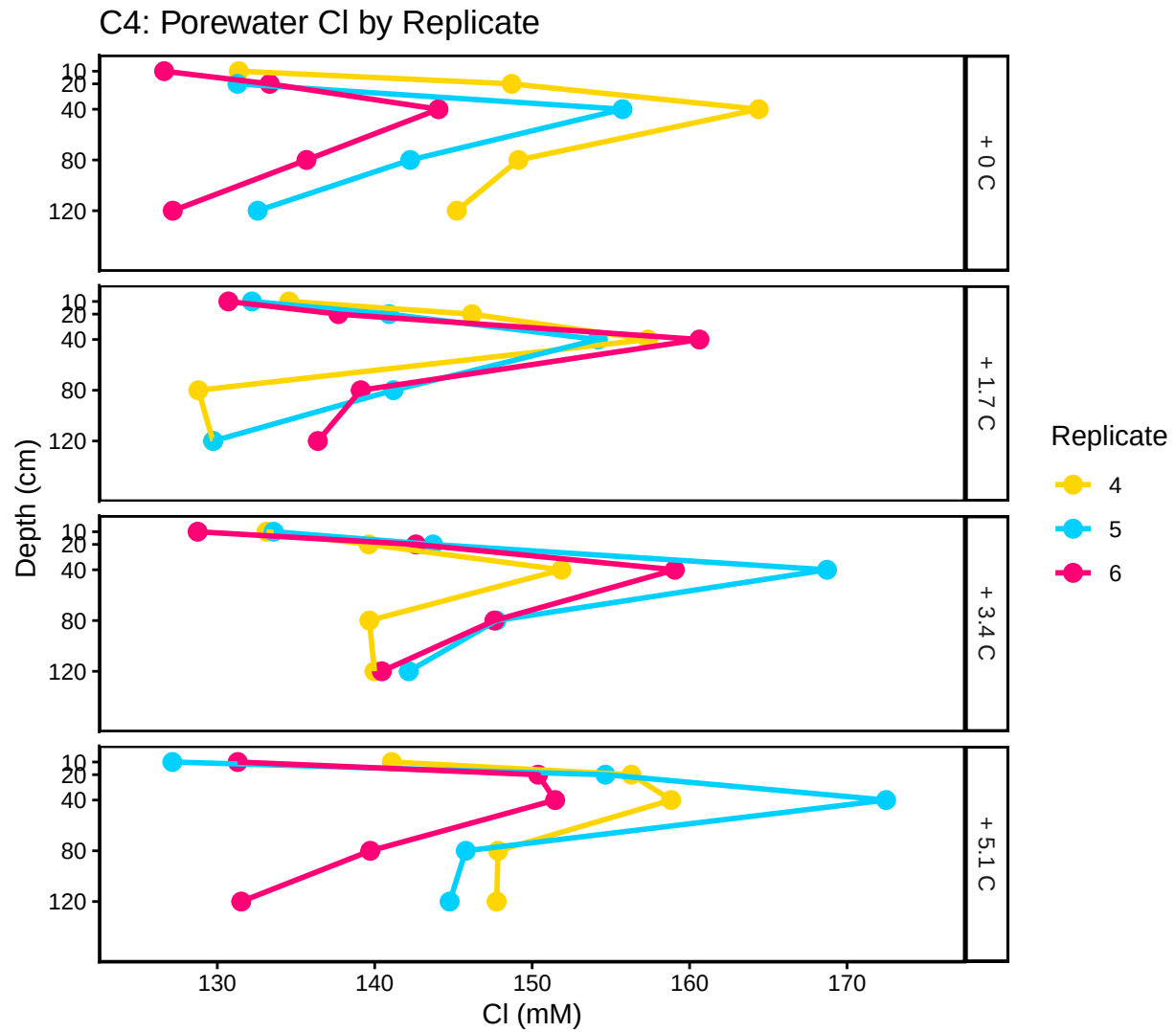


C3: Porewater Cl by Replicate



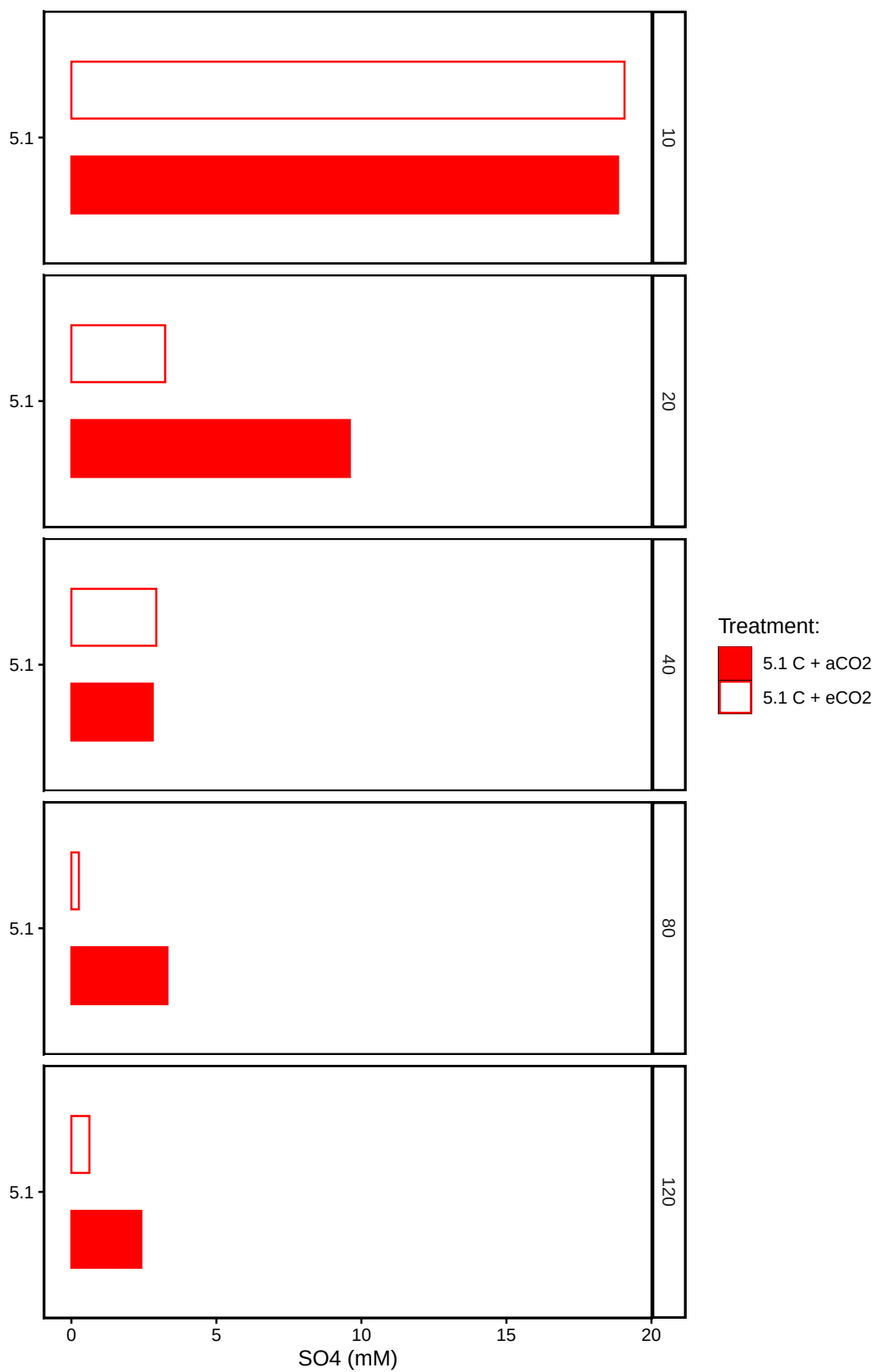
```
## Warning: Removed 1 row containing missing values or values outside the scale range
## ('geom_point()').
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## ('geom_path()').
```

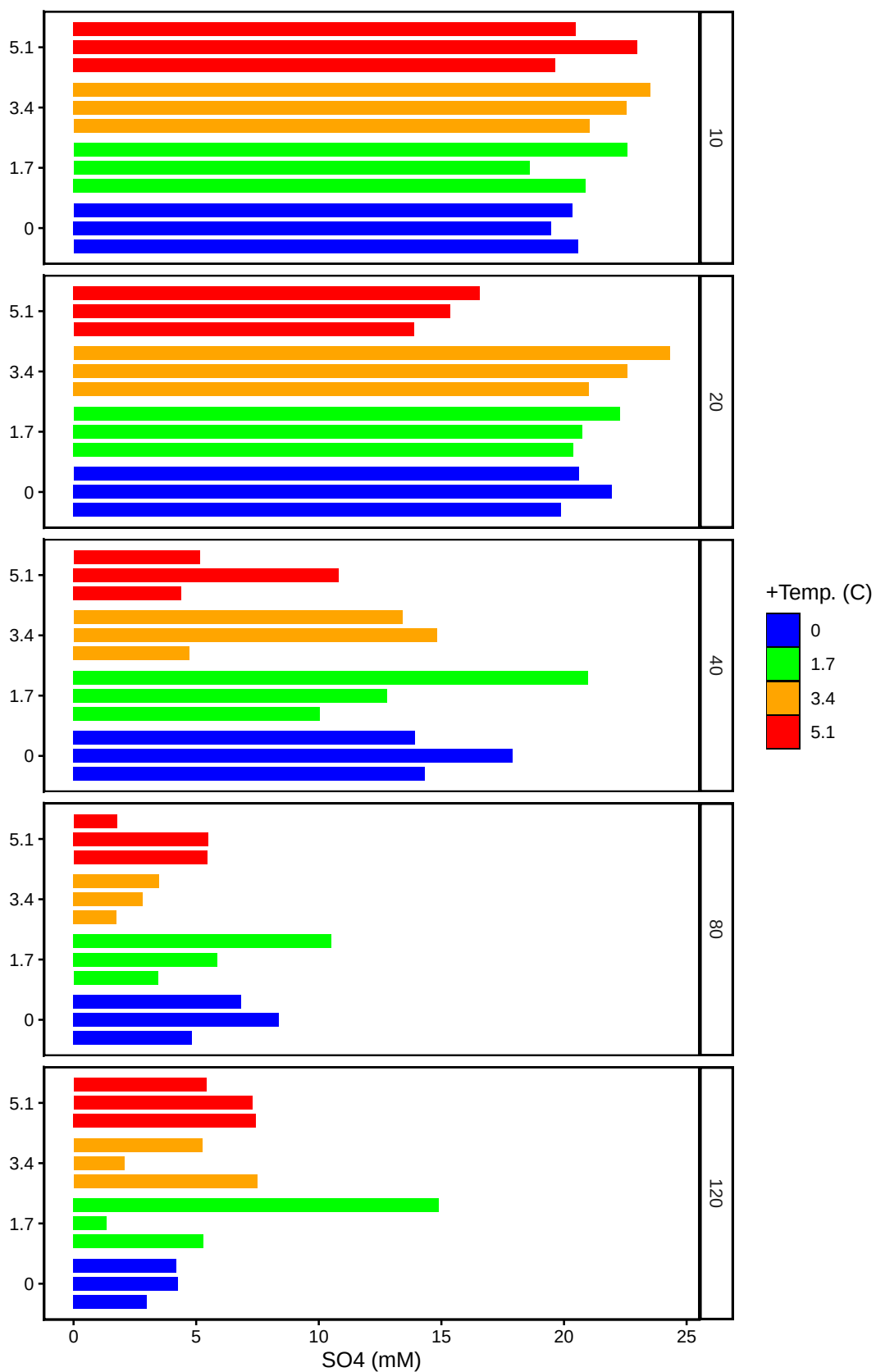


```
## Visualize Data - S04
```

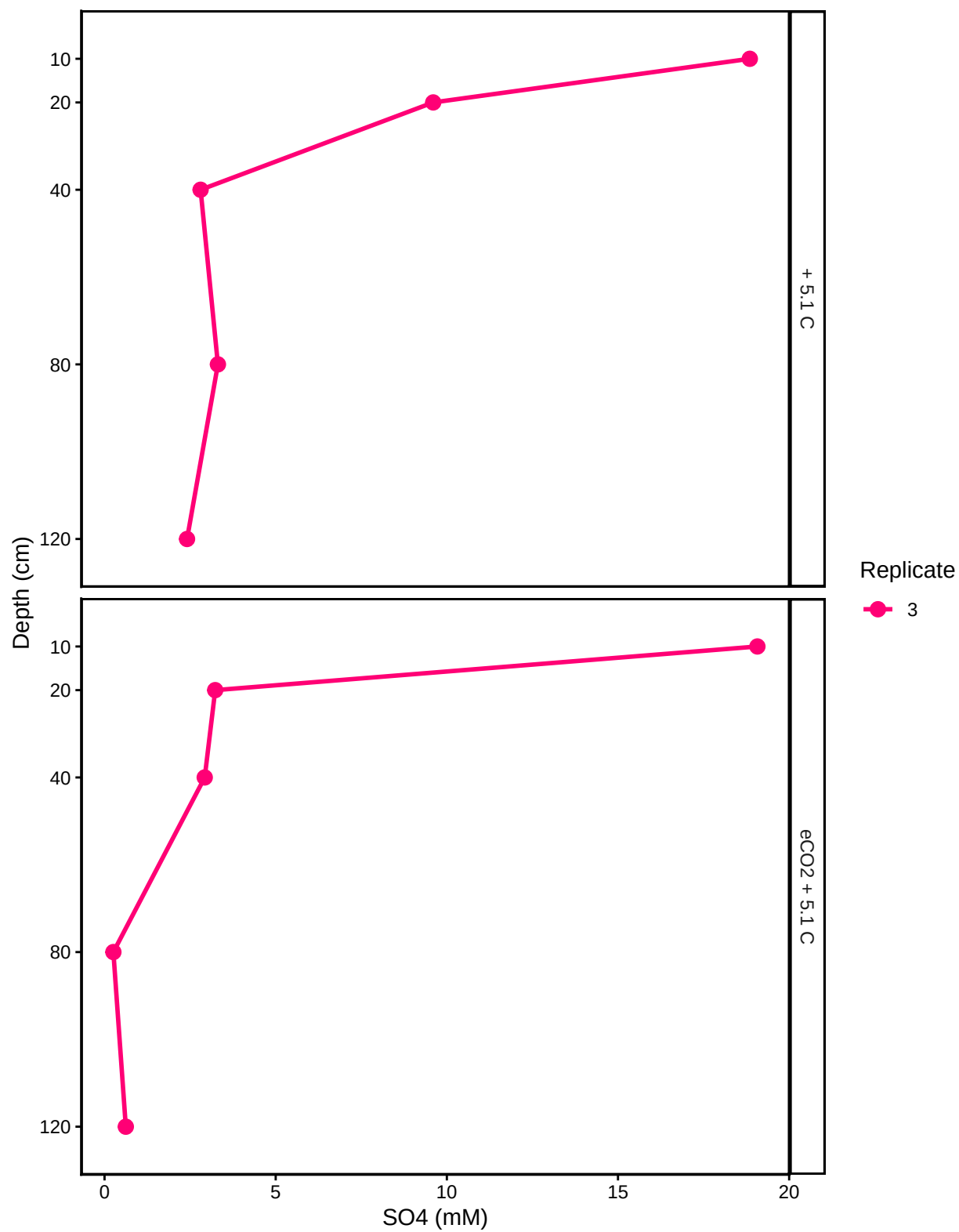
C3: Porewater SO₄

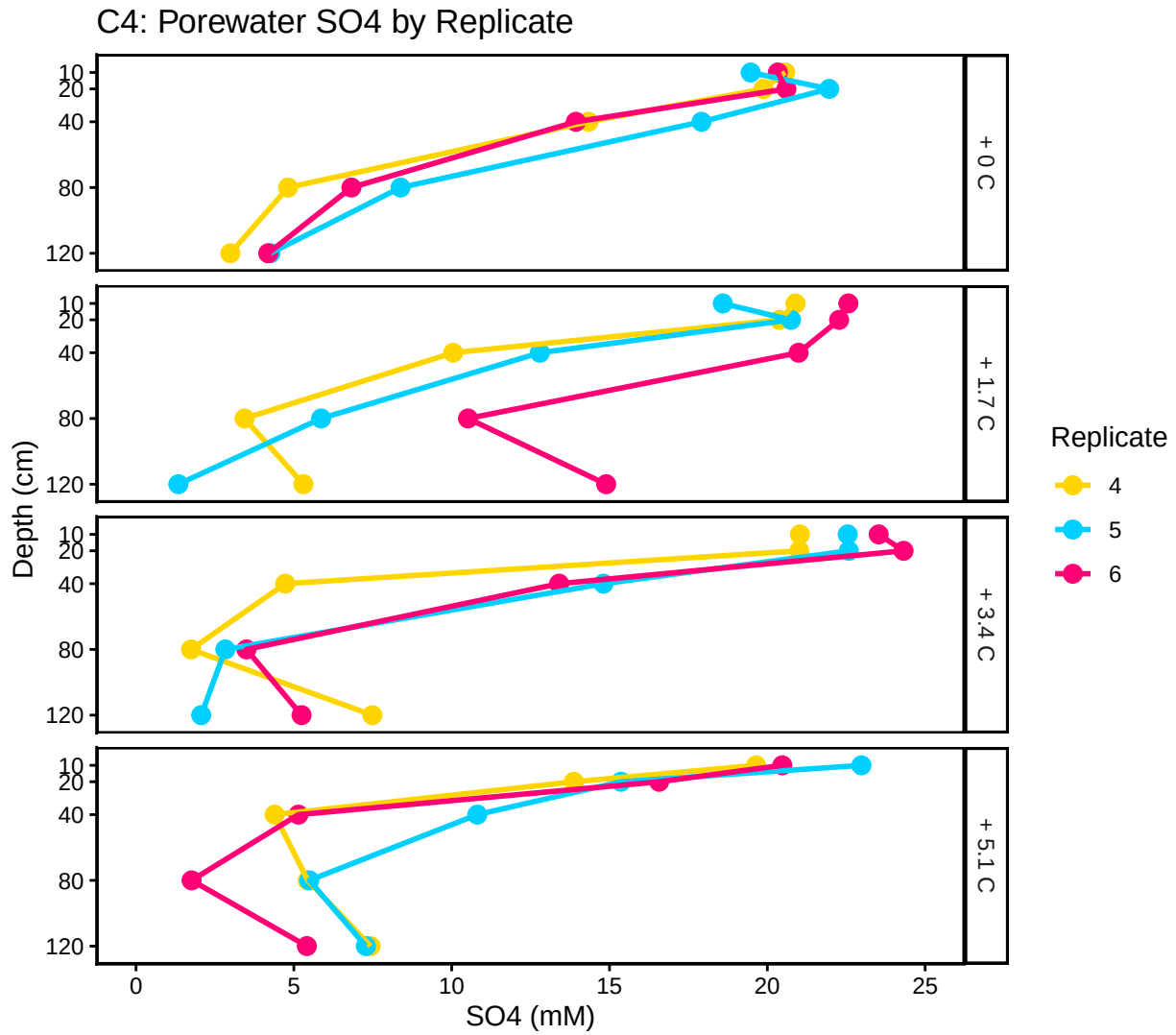


C4: Porewater SO4



C3: Porewater SO₄ by Replicate





0.11 Export Processed Data

#end